

Software Product Project Report

Enterprise Business Operating System (BizOS) & Integrated Multi-Tenant Marketplace

Platform: Office Connect / BizOS
Classification: Enterprise SaaS Architecture
Release Date: September 2026

COMPREHENSIVE SPEC V1.0

6-in-1

UNIFIED CORE MODULES

Multi-Tenant

LOGICAL ISOLATION (RLS)

WebRTC

EMBEDDED VIDEO HUB

API-First

EXTENSIBLE INTEGRATION

1 Project Overview

The Core Concept: The platform is an integrated, multi-tenant Business Operating System (BizOS) designed for startups, SMEs, mid-market organizations, and global enterprises. It consolidates disparate business operations—workforce management, sales pipelines, multi-warehouse inventory, embedded WebRTC video meetings, and multi-vendor marketplace trading—into a unified cloud application.

Modularity & Subscriptions: Organizations are not burdened with software bloat. The system features a modular architecture where organizations subscribe to the platform and dynamically activate modules matching their operating scale. As the organization grows, additional modules, custom enterprise workflows, and third-party integrations are provisioned without data migrations or platform transitions.

2 Main SaaS Modules

2.1 User and Organization Onboarding

- **Company Registration & Entity Modeling:** Self-service portal supporting diverse corporate entities (Private Limited, Public Limited, LLP, Sole Proprietorship, Non-Profit).
- **Identity & KYC Verification:** Email magic links, mobile SMS OTP, automated tax format validation (GSTIN, PAN, VAT, Tax ID), and corporate document uploads.
- **Organizational Hierarchy Creation:** Multi-branch structures, department hierarchies (Sales, Engineering, HR, Ops), and team reporting structures.
- **Bulk User Provisioning:** Invitation link generation and CSV/XLSX bulk user ingestion with pre-assigned RBAC roles and manager assignments.
- **Initial Configuration Wizard:** Guided onboarding wizard adapting configuration to the company's industry vertical in under 15 minutes.

2.2 CRM – Customer Relationship Management

- **Lead & Contact Management:** Omnichannel lead capture from web forms, WhatsApp, marketplace inquiries, and email integrations.
- **Sales Pipeline & Deal Stages:** Visual Kanban deal pipeline with stages: Lead In -> Qualified -> Demo/Meeting -> Proposal -> Negotiation -> Won/Lost.
- **Activity Ledger:** 360-degree timeline tracking calls, meeting logs, email threads, notes, and task assignments.
- **Automated Lead Assignment:** Round-robin, regional territory, and capacity-based automatic lead routing.
- **Sales Forecasting & Performance:** Weighted deal probability, conversion rate metrics, and individual sales rep quota pacing.

2.3 HRM – Human Resource Management

- **Employee Records:** Centralized digital employee dossier with employment contracts, statutory IDs, and emergency contacts.
- **Attendance & Shift Scheduling:** Web check-in, geofenced mobile clock-in, biometric machine API sync, and dynamic shift rotations.

- **Leave & Holiday Administration:** Configurable leave policies (Casual, Sick, Earned, Unpaid), accruals, multi-tier approvals, and regional holiday calendars.
- **Payroll Integration:** Automated calculation of base pay, statutory deductions (TDS, PF, ESI, PT), reimbursements, and bank payout batches.
- **Performance & Appraisals:** OKR and KPI tracking frameworks, 360-degree reviews, and quarterly appraisal scorecards.
- **Offboarding & Exit Workflow:** Automated asset return tracking, exit interview logging, and final settlement calculation.

2.4 Inventory Management

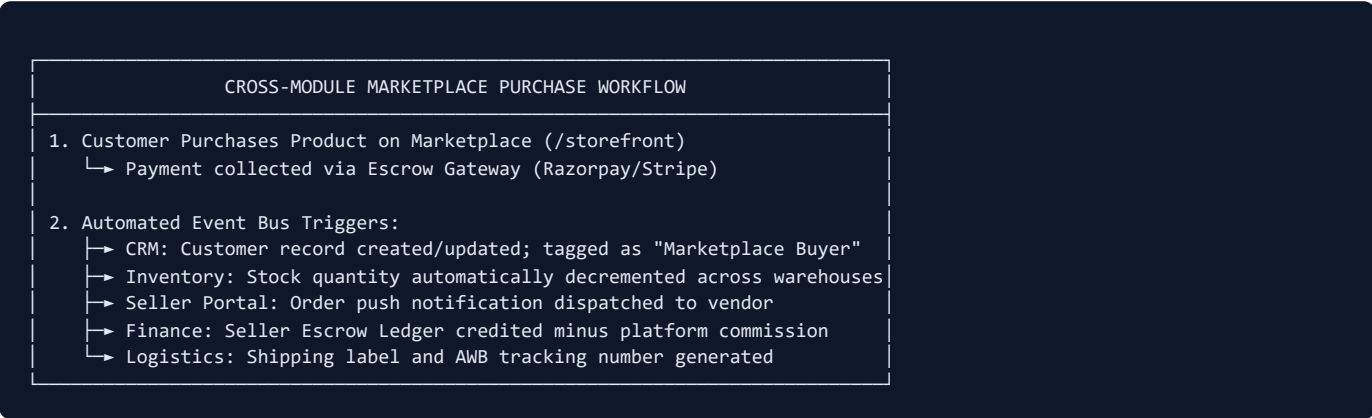
- **Product & SKU Management:** Variant tracking (size, color, material), barcodes, QR codes, and multi-currency pricing.
- **Multi-Warehouse Architecture:** Stock visibility across multiple physical warehouses, retail locations, and transit hubs.
- **Stock Movements:** Goods Receipt Notes (Stock-In), Order Dispatch (Stock-Out), Inter-Warehouse Transfers, and Shrinkage Adjustments.
- **Procurement & Suppliers:** Supplier directories, automated Purchase Order (PO) generation, and supplier invoice reconciliation.
- **Safety Stock Alerts:** Automated notifications triggered when inventory reaches minimum reorder thresholds.

2.5 Video Calling and Communication Hub

- **Embedded WebRTC Infrastructure:** Low-latency HD video and audio sessions directly inside CRM leads, HR interviews, and support desks.
- **One-to-One & Group Meetings:** Instant peer-to-peer calls, scheduled client demos, and multi-participant team standups.
- **In-Meeting Collaboration:** High-framerate desktop/tab screen sharing, persistent chat, hand-raising, and participant muting.
- **Meeting Logs & Records:** Attendance timestamps, meeting duration telemetry, and optional cloud recording integrations.
- **Zero-Download Access:** Native browser execution across modern web browsers without requiring third-party software installation.

3 Marketplace Module

- The platform contains an integrated B2B/B2C marketplace that enables organizations to act as buyers, sellers, or network operators.
- **Merchant Portal & Onboarding:** Merchant registration with statutory KYB checks (Business Registration, Bank Penny Drop, Tax ID).
 - **Catalog Management:** Self-service SKU listings, image galleries, technical specifications, and category taxonomy assignment.
 - **Storefront & Search:** Responsive public marketplace with full-text fuzzy search, faceted filtering, and promotional banners.
 - **Escrow Settlement Ledger:** Automated category commission deductions (e.g., 8%) with 7-day escrow holdbacks before direct bank disbursement.
 - **Unified Order Orchestration:** Cross-module automation between marketplace sales, CRM, and inventory.



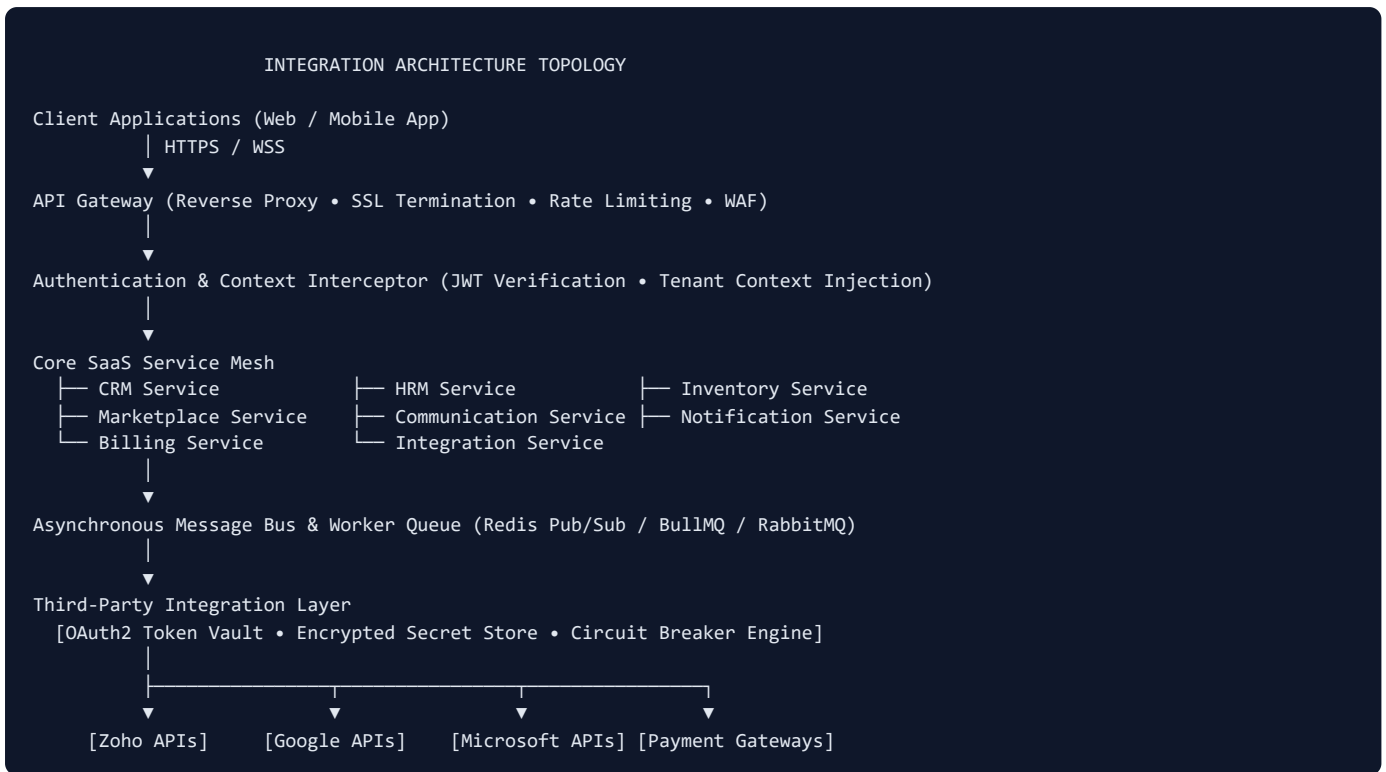
4 Third-Party Integrations

Organizations frequently maintain existing software investments. The platform uses an **API-First Extensibility Layer** so organizations can integrate external applications rather than being forced into wholesale migrations.

INTEGRATION CATEGORY	SUPPORTED THIRD-PARTY SYSTEMS & CAPABILITIES
Zoho Ecosystem	Zoho CRM, Zoho Books (Invoices & Credit Notes), Zoho People, Zoho Inventory, Zoho Desk, Zoho Projects
Google Workspace	Google Calendar synchronization, two-way Gmail email thread logging into CRM, Google Drive, Google Meet
Microsoft 365	Outlook Calendar, Microsoft Teams meetings and presence status, OneDrive document linking, Entra ID (Azure AD) SSO
Accounting & ERP	Tally Prime, QuickBooks Online, Xero Books, Zoho Books daily ledger and tax batches
Omnichannel Comms	WhatsApp Business API (transactional notifications), Twilio SMS, Amazon SES, SendGrid
Payment Gateways	Razorpay & Razorpay Route (Virtual Escrow), Stripe Connect, PayPal, Cashfree, PayU
E-Commerce Platforms	Shopify, WooCommerce, Amazon SP-API, and custom webhook-driven stores

Enterprise Extensibility Guarantee: If a customer requires an integration that is not currently available out of the box, the integration can be developed and added based on the customer's specific requirements, provided the third-party service offers a supported REST, GraphQL, SOAP, or Webhook API.

5 Integration Architecture



Technical Highlights: 1. *OAuth 2.0 Management:* Token refresh cycles and encrypted secret storage using AES-256. 2. *Circuit Breaker Pattern:* Protects platform compute resources if external APIs experience downtime. 3. *Audit Logging:* Inbound webhook payloads, outbound API requests, and response status codes are immutably logged for debugging and compliance.

6 Multi-Tenant SaaS Architecture

The platform utilizes a **Shared Infrastructure with Logical Tenant Isolation** architecture, delivering resource efficiency alongside strict data security.

- **Tenant Identification (`organization\_id`)**: Every database table contains a mandatory, foreign-keyed `organization\_id` column.
- **PostgreSQL Row-Level Security (RLS)**: Enforced at the database engine level. Queries execute within a scoped tenant context, preventing cross-tenant data leaks even in the event of an application-layer bug.
- **Isolated File Storage**: Tenant document assets are partitioned in S3 object storage paths (`/storage/tenants/{organization\_id}/...`) accessible only via temporary, presigned, scoped URLs.
- **Tenant-Specific Configurations**: Modular feature toggles, branding, currency preferences, and custom pipeline stages are loaded dynamically per tenant.

7 Roles and Permissions (RBAC)

ROLE	SCOPE	PERMISSIONS & ACCESS BOUNDARIES
SaaS Super Admin	Global Platform	Full control across all organizations, billing plans, tenant provisioning, system logs, and global feature flags.
Organization Admin	Single Organization	Full control over tenant users, department setup, billing subscription, module activations, and tenant settings.
Sales Manager	Sales Department	Full CRM deal pipeline, sales reports, team lead assignments. Read-only inventory access.
CRM Sales Executive	Individual / Assigned	Manage assigned leads, schedule customer meetings, update deal stages. No access to other reps' unassigned deals.
HR Manager	HR Department	Full employee records, attendance rosters, leave approvals, payroll processing, performance appraisals.
Inventory Manager	Warehouse / Supply	Full SKU creation, stock-in (GRN), inter-warehouse transfers, supplier orders, minimum stock alerts.
Marketplace Vendor	Vendor Portal	Catalog management, product uploads, order dispatch, packaging slips, escrow payout ledger.
Standard Employee	Self-Service	View personal profile, clock-in/out, apply for leaves, access internal announcements, join video meetings.
Customer / Buyer	Storefront / Orders	Public catalog browsing, shopping cart, checkout, order tracking timeline, support tickets.

8 Subscription Management

PLAN TIER	PRICING MODEL	INCLUDED CAPABILITIES & MODULES
Basic Plan	₹2,999 / month (or annual)	CRM Core + Basic HRM (up to 25 employees), Single Branch, Email Support, 5 GB Storage
Professional Plan	₹7,999 / month (or annual)	Full CRM + Full HRM + Inventory Core + WebRTC Video (up to 10 pax), 50 GB Storage
Business Plan	₹18,999 / month (or annual)	Full CRM + Full HRM + Multi-Warehouse Inventory + Vendor Marketplace + Integrations Hub, 250 GB Storage
Enterprise Plan	Custom Contract	All Modules + Custom Integrations + Dedicated Infrastructure Node + 99.99% SLA + Unlimited Storage

9 Super Admin Panel (Platform Governance)

- **Tenant Management:** Provision, suspend, upgrade, or archive customer organizations with real-time resource tracking.
- **Commercial Governance:** Platform-wide MRR/ARR tracking, payment gateway collection logs, commission split auditing.
- **Modular Feature Flags:** Globally activate or canary-test new platform modules for select organizations.
- **Security Monitoring:** Global authentication failure tracking, rate limit violation alerts, and administrative session audits.

10 Organization Dashboard

A unified executive dashboard providing leadership with real-time operational visibility:

- **Workforce Metrics:** Active headcount, attendance percentage, pending leave requests.
- **Sales Pipeline:** Active pipeline value, deals closing this month, top-performing sales executives.
- **Supply Chain Status:** Real-time inventory valuation, low-stock alerts, pending supplier purchase orders.
- **Commerce Performance:** Marketplace Gross Merchandise Value (GMV), active seller orders, escrow balances.
- **Action Center:** Pending approvals, scheduled customer demos, and operational alerts.

11 Workflow Automation

REPRESENTATIVE AUTOMATED BUSINESS LOGIC

A. Lead Qualification:

[New Lead Ingested] → [Lead Value > ₹100,000?]
└─ (Yes) Assign to Senior Director & Schedule Demo
└─ (No) Assign to Round-Robin SDR & Send WhatsApp Deck

B. Inventory Replenishment:

[Stock Drops Below Minimum Threshold] → [Generate Automated Supplier Purchase Order]
└─ [Alert Inventory Head & Warehouse Clerk]

C. Employee Lifecycle:

[HR Admin Approves New Hire] → [Generate Employee ID & Welcome Email]
└─ [Provision Department Access & Assign Manager]
└─ [Schedule Day-1 Video Check-in Meeting]

12 Omnichannel Notification System

- **Supported Channels:** In-App Bell notifications, Web Push (Chrome/Edge/Firefox), Transactional Email (Amazon SES), WhatsApp Business API, and SMS Gateways.
- **Configurable Rules:** Users configure notification preferences by channel and urgency level.
- **Reliability:** Failed notification delivery automatically falls back (e.g., WhatsApp delivery failure triggers SMS fallback).

13 Reporting and Analytics

DOMAIN	KEY METRICS & ANALYTICS DASHBOARDS
CRM Analytics	Lead-to-deal conversion rates, sales cycle length, Customer Acquisition Cost (CAC), Lifetime Value (LTV), rep quota pacing
HRM Analytics	Headcount growth, absenteeism heatmaps, department payroll distribution, attrition rates, employee review scores
Inventory Analytics	Inventory turnover ratio, dead stock analysis, carrying costs, SKU stockout frequencies, supplier lead-time variance
Marketplace Analytics	Gross Merchandise Value (GMV), platform commission income, seller fulfillment times, return and refund rates
Platform SaaS BI	Monthly Recurring Revenue (MRR), churn rate, Customer Acquisition Cost, Active Organizations, API usage volume

14 Enterprise Security & Compliance

- **Encryption:** TLS 1.3 in transit; AES-256 block encryption at rest for databases and file storage.
- **Authentication Security:** Bcrypt password hashing (cost factor 12), mandatory MFA/TOTP for administrative accounts, session timeouts.
- **OWASP Defense:** Strict parameterized queries via Prisma ORM (SQLi-proof), automatic JSX output escaping (XSS-proof), CSRF protection.
- **Rate Limiting & DDoS:** Token-bucket rate limiting via Redis, Web Application Firewall (WAF) integration.
- **Audit Logging:** Immutable audit records capturing user logins, financial transactions, and document exports.

15 Recommended Technology Stack

LAYER	PRODUCTION TECHNOLOGY CHOICES & RATIONALE
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Frontend Application	Next.js 16 (React 19, Turbopack, App Router) with Tailwind CSS for high performance, server-side rendering, and responsive UI.
Backend API	Node.js (TypeScript) with Express / Fastify for event-driven, asynchronous throughput. Optional Java Spring Boot for high-volume financial ledgers.
Database Layer	PostgreSQL 16 with Prisma ORM for robust relational integrity, JSONB support, and Row-Level Security (RLS).
In-Memory Cache	Redis 7.2 for session caching, rate limiting, and BullMQ background task processing.
Real-Time Media	WebRTC with Mediasoup / LiveKit SFU for peer-to-peer and multi-party video conferencing.
File Storage	AWS S3 / Cloudflare R2 / MinIO for secure, presigned S3-compatible object storage.
Infrastructure	Docker, Nginx 1.24 Reverse Proxy, Hostinger Cloud VPS / AWS Kubernetes (EKS).

16 API Architecture

Adheres strictly to RESTful OpenAPI 3.0 conventions with standardized JSON envelopes:

```
API ROUTE NAMESPACE:
/api/v1/auth           -> Session management, OTP verification, token refresh
/api/v1/organizations  -> Tenant provisioning, corporate profile, settings
/api/v1/users          -> User management, invitations, password resets
/api/v1/crm            -> Leads, deals, sales pipeline, contacts, timeline
/api/v1/hrm            -> Employees, attendance, shifts, leave applications, payroll
/api/v1/inventory      -> Products, SKUs, warehouses, stock-in/out, adjustments
/api/v1/marketplace     -> Vendors, public catalog, cart, multi-vendor orders
/api/v1/communication  -> WebRTC room creation, meeting tokens, recordings
/api/v1/integrations    -> OAuth handshakes, partner webhooks, sync tasks
/api/v1/subscriptions   -> Plan management, usage metrics, invoice downloads
```

17 Database Entity Architecture

Core database entities maintain strict referential integrity linked by the mandatory multi-tenancy key `organization_id`:

- `organizations`: Root tenant record (name, domain, subscription tier, status).
- `users & roles`: System credentials, RBAC policy bindings, department links.
- `hrm_employees & departments`: Workforce records, attendance logs, leave balances, payroll slips.
- `crm_leads & crm_deals`: Sales prospects, deal stages, assigned sales reps, communication logs.
- `inventory_products & warehouses`: Master SKU records, stock quantities, warehouse bin allocations.
- `marketplace_orders & escrow_ledger`: Transactions, vendor payout balances, commission holdbacks.
- `audit_logs`: Immutable records of administrative and security events.

18 Scalability & Resilience

- **Horizontal Stateless Scaling**: Node.js application instances run statelessly behind Nginx / AWS ALB load balancers, scaling dynamically with traffic.
- **Database Read/Write Splitting**: Master PostgreSQL database handles write transactions, while read replicas handle dashboard analytics and catalog queries.
- **Asynchronous Task Offloading**: CPU-heavy tasks (report generation, video encoding, email dispatches) are processed asynchronously via Redis worker queues.

19 Strategic Product Advantages

DIMENSION	TRADITIONAL FRAGMENTED SOFTWARE	THE OFFICE CONNECT UNIFIED PLATFORM
Data Synchronization	Fragile API sync tools with frequent sync delays	Native zero-latency database consistency across all modules
Subscription Costs	Compounding per-user fees across 5+ independent tools	Consolidated modular subscription saving 40% to 60% in software spend
User Administration	Manual provisioning across multiple vendor systems	Single-click centralized user onboarding and offboarding
Operational Workflows	Isolated silos requiring human copy-paste work	Automated event triggers across CRM, HR, Inventory, and Marketplace

20 Unique Selling Proposition (USP)

"One unified platform to run your entire business:
manage your people, engage your customers, track your inventory,
conduct face-to-face communication, and monetize your marketplace."

21 Future Scope & Product Roadmap

- **Autonomous AI Assistants:** AI Sales Assistant (automated email follow-ups), AI HR Assistant (leave and policy guidance), Predictive Stock Forecasting.
- **Extended ERP Suite:** Double-entry General Ledger accounting, Fixed Asset Management, Enterprise Project Management with Gantt charts.
- **Hardware & Mobile Expansion:** Native iOS & Android apps, IoT warehouse barcode scanner integration, and digital PKI e-signatures.

22 Development Phases

PHASE	FOCUS AREA	DELIVERED CAPABILITIES
Phase 1	Foundations	Multi-tenant SaaS engine, PostgreSQL RLS schema, auth, onboarding wizard, subscriptions
Phase 2	CRM & HRM	Lead management, deal stages, employee directory, attendance tracking, leave workflows
Phase 3	Inventory	SKU tracking, multi-warehouse support, stock-in/out, reorder alerts, supplier POs
Phase 4	Communication	Embedded WebRTC video meetings, screen sharing, multi-channel notification engine
Phase 5	Marketplace	Vendor onboarding, catalog management, public storefront, escrow payout ledger
Phase 6	Integrations	Standard connectors for Zoho, Google, Microsoft, Tally, payment gateways, custom webhooks
Phase 7	Automation	Cross-module workflow automation engine, executive BI dashboards, scheduled reporting
Phase 8	Scale & AI	Autonomous AI assistants, SOC 2 compliance readiness, global high-availability clustering

23 Final Project Conclusion

This platform transitions organizations away from disconnected point applications toward a cohesive **Enterprise Business Operating System**. By unifying workforce management, customer relationships, supply chain inventory, real-time video communication, and marketplace commerce under a single secure multi-tenant architecture, the platform eliminates operational friction, reduces overall software expenditure, and provides modern organizations with the digital foundation required for sustainable growth.